CO1 ASSIGNMENT

C++ Programming – Corrected Programs, Sample Inputs and Outputs

Name: _____ Register No.: _____

Department: _____ Subject: _____

QUESTION 1: Find the Largest of Three Numbers

Aim: To find and display the largest among three given integers.

Corrected C++ Program:

```
#include <iostream>
using namespace std;

int main() {
    int a, b, c;
    cin >> a >> b >> c;

    if (a > b && a > c)
        cout << a;
    else if (b > a && b > c)
        cout << b;
    else
        cout << c;

    return 0;
}
```

Sample Input	Expected Output
15 28 20	28

Explanation: The program compares each number with the other two. Since 28 is greater than both 15 and 20, the output is 28. The combined conditions avoid the incorrect nested-else result.

QUESTION 2: Sum of Numbers Divisible by 3 from 1 to n

Aim: To calculate the sum of all numbers divisible by 3 from 1 through n.

Corrected C++ Program:

```
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;
    cin >> n;

    for (int i = 1; i <= n; i++) {
        if (i % 3 == 0)
            sum += i;
    }

    cout << sum;
    return 0;
}
```

Sample Input	Expected Output
12	30

Explanation: The loop checks every integer from 1 to 12. The condition i % 3 == 0 selects 3, 6, 9 and 12. The += operator accumulates them: 3 + 6 + 9 + 12 = 30.

QUESTION 3: Calculate a Raised to the Power b

Aim: To calculate a^b using a loop, where b is a non-negative integer.

Corrected C++ Program:

```
#include <iostream>
using namespace std;

int main() {
    int a, b;
    int result = 1;
    cin >> a >> b;

    for (int i = 1; i <= b; i++) {
        result *= a;
    }

    cout << result;
    return 0;
}
```

Sample Input	Expected Output
3 4	81

Explanation: The loop runs b times and multiplies result by a on every iteration. Therefore, 3 × 3 × 3 × 3 = 81. When b is 0, the initial result 1 is printed.

QUESTION 4: Calculate Simple Interest

Aim: To calculate simple interest using principal, rate and time.

Corrected C++ Program:

```
#include <iostream>
using namespace std;

int main() {
    double principal, rate, time, interest;
    cin >> principal >> rate >> time;

    interest = (principal * rate * time) / 100;

    cout << interest;
    return 0;
}
```

Sample Input	Expected Output
5000 6 2	600

Explanation: The inputs are read in the order principal, rate and time. Using SI = (P × R × T) / 100 gives (5000 × 6 × 2) / 100 = 600.

Conclusion

The four C++ programs have been corrected and tested against the sample inputs shown. They demonstrate conditional statements, loops, accumulation, exponentiation and arithmetic calculations.